

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027
Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clients**, building REST integrations that connect the Mayden iaptus platform and process **10,000+ monthly submissions**; contributed to **\$340K+** in revenue.
- Architected a full-stack data platform (React, Node.js, MongoDB) used daily by the AI team, cutting manual work by **100+ hours/month** and lifting throughput by **60%**.
- Implemented backend reliability and security controls, AWS KMS encryption, auth hardening, and rate limiting, and remediated **20+ VAPT findings**.
- Served as primary technical POC, debugged production incidents to root cause, and piloted a COO-sponsored GitHub Copilot experiment.

Cario Growth Services, Backend Developer Intern | *Chennai, India* Nov 2022 – May 2023

- Built and maintained **20+ RESTful APIs** (Node.js, Fastify, **PostgreSQL/SQL**) with object-oriented design, input validation, and clear contracts connecting systems in production.
- Migrated core repositories from JavaScript to TypeScript and shipped a production feature running a **4-bit quantized Llama-7B** model via llama.cpp with async queued generation.

Freelance / Open Source, Full-Stack Engineer | *Remote* Jul 2022 – Aug 2023

- Built and deployed **Underdogs Fitness** (Website), a production MERN platform with REST APIs, a multi-branch data model, and cron automation, independently end to end; live **3+ years**.

Projects

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a Python/FastAPI back-end with **REST API endpoints** that integrate multiple data sources (web search and multiple LLM providers) via typed retrieval planning, deterministic extraction, and provider routing; hardened with **150+ tests** and **Docker** deployment.

Spark ETL Performance Optimization (GitHub) PySpark, Spark UI, SQL, NYC TLC Dataset

- Built **distributed data pipelines** processing a **~1.4B-row** dataset with broadcast joins, skew-join handling, partitioning, column pruning, and a data-quality filter, cutting end-to-end runtime **~25%**, profiled via Spark UI.

cf-ai-research-scout – Edge-Native AI Assistant (GitHub) TypeScript, Cloudflare Workers, Durable Objects, D1

- Built a distributed, **event-driven** service in TypeScript with durable workflows that survive process restarts and tool failures, streaming results to the client over WebSocket in real time.

Publications

- “Alzheimer’s Disease Classification using Transfer Learning,” IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages: JavaScript, TypeScript, Python, SQL, Java, C++, HTML/CSS, Bash

Backend & APIs: REST APIs, Node.js, Express, FastAPI, object-oriented design, integration testing

Data & Distributed: Spark, data pipelines, Kafka, event-driven, distributed systems, ETL

Frontend & DevOps: React, Next.js, Docker, Kubernetes, Git, GitHub Actions, CI/CD

Cloud & Platforms: AWS, GCP, Azure, Linux, MongoDB Atlas, PostgreSQL

Achievements

- Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.